

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient watches. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each hand separately by evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide a score on the last line.

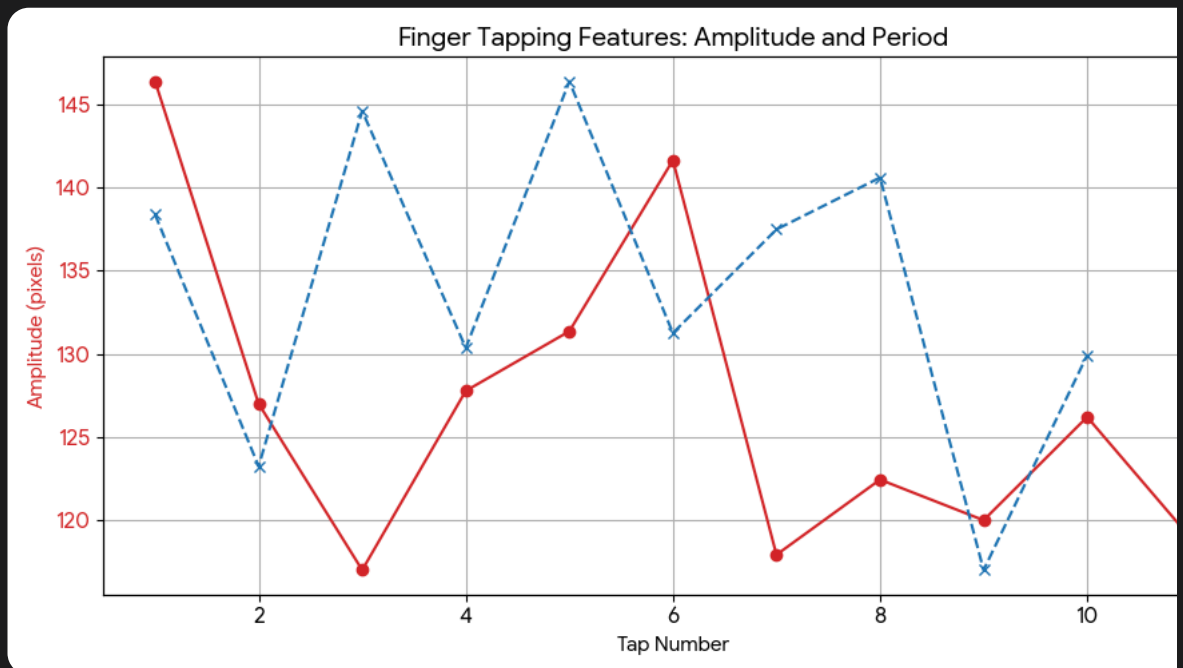
EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]

Tap amplitudes (pixels): [146.40013661195812, 127.00393694685216, 117.00427342623003, 127.781062759706, 131.36590120727678, 141.65097952361643, 117.92370414806346, 122.44182292011173, 120.0, 126.21014222319, 118.92854997854805]





Based on the extracted features provided, here is the analysis according to the MDS-UPDRS Item 3.4 criteria:

1. Amplitude Analysis:

- **Data:** [146.4, 127.0, 117.0, 127.8, 131.4, 141.7, 117.9, 122.4, 120.0, 126.2, 118.9]
- **Observation:** The subject starts with a maximum amplitude of ~146 pixels. There is an immediate drop after the first tap (~117). However, the amplitude recovers midway through the sequence (Tap 6 reaches ~141.7 pixels, close to the baseline) before dropping again.
- **Assessment:** While the amplitude drops after the first tap, the significant recovery at the 6th tap contradicts a consistent "decrementing" sequence effect (fatigue) typically associated with Score 3 or 4. The pattern is better described as variable or mild inability to maintain amplitude.

2. Rhythm and Speed (Periods) Analysis:

- **Data:** [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]
- **Observation:** The tap periods exhibit a significant oscillating or "sawtooth" pattern (Long-Short-Long-Short). For example, jumps from 155 to 203 (+48 frames) and back to 171 (-32 frames).
- **Assessment:** This indicates notable dysrhythmia. If we define the faster taps (~155-170 frames) as the intended pace periods (189, 203, 207, 187, 194) represent hesitations or interruptions in the rhythm. There are exactly 5 such hesitations in the 10-period sequence.

3. Speed (Slowing):

- **Trend:** There is no progressive slowing trend (periods do not get consistently larger; in fact, the slope is slightly negative).

Scoring Conclusion:

- **Score 0 (Normal):** Incorrect due to obvious rhythm and amplitude issues.
- **Score 1 (Slight):** Incorrect. The subject has more than "one or two" interruptions.
- **Score 2 (Mild):** The criteria state "3 to 5 interruptions during tapping." The analysis identifies approximately 5 distinct hesitations/interruptions in the rhythm. Additionally, the amplitude variability fits a mild profile better than a severe progressive decrement.
- **Score 3 (Moderate):** Incorrect. The subject does not have "more than 5 interruptions" or a "freeze" (arrest). While an early drop occurs, the recovery at Tap 6 prevents it from meeting the strict definition of a continuous pathological decrement.

The performance fits the criteria for **Mild** impairment primarily due to the number of rhythm interruptions.

